NAME: R. JAGAN

REG NO: 192321102

COURSE CODE: ITA0609 – MACHINE LEARNING

C04 – AT3

COMPARATIVE ANALYSIS OF CASE-BASED LEARNING, LOCALLY WEIGHTED REGRESSION, AND RBF NETWORKS

1. Case-Based Learning vs. Locally Weighted Regression

Both Case-Based Learning (CBL) and Locally Weighted Regression (LWR) are instance-based learning methods. They store training examples and use the similarity between a new query and previous examples during prediction. LWR creates a local approximation around each query rather than learning one global model during training.angom.myweb.uwindsor+1

Practical Example: Recommendation Systems

Consider a movie recommendation system. The system stores information such as:

- User age and language.
- Previously watched movies.
- Movie genre and release year.
- User ratings and watch time.
- Whether the user liked or skipped a recommendation.

The system must estimate how much a user will like a new movie.

Case-Based Learning

Case-Based Learning stores previous situations as cases. A case may contain:

- User profile.
- Movie characteristics.
- Previous recommendation.
- User response.
- Final outcome.

When a new recommendation request arrives, the system retrieves similar past cases and adapts their solutions.

For example, suppose a new user enjoys Tamil action movies and has highly rated several films starring a particular actor. The system searches for previous users with similar preferences and recommends movies that were successful for those users.

Locally Weighted Regression

Locally Weighted Regression predicts a numerical value, such as a rating or probability of watching. It gives greater importance to training examples that are close to the current user-movie pair.

For example, if a user has similar preferences to users who gave a particular movie ratings of 4, 4.5, and 5, those nearby examples receive high weights. Dissimilar users receive smaller weights. LWR then fits a local regression model to estimate the user's expected rating.

LWR uses nearby examples to construct a local model only when a query is received.web.cs.hacettepe.edu

Comparison of the Two Methods

Criterion	**Case-Based Learning**	**Locally Weighted Regression**
Representation	Stores complete cases containing context, problem, solution, and outcome.	Stores feature vectors and numerical target values.
Prediction	Retrieves similar cases and adapts their previous solutions.	Assigns distance-based weights and fits a local regression model.
Output	Can provide a recommendation and explanation.	Usually produces a numerical prediction or score.
Best suited for	Context-rich, symbolic, and explanation-oriented tasks.	Continuous numerical prediction.
Recommendation example	"Users with similar interests liked this movie."	"This user is predicted to give the movie a rating of 4.4."
Learning style	Lazy or memory-based.	Lazy or memory-based.

Handling New Data

Case-Based Learning

CBL handles new information by adding new cases to its case library. Once a user rates a movie, that interaction can immediately become a case for future recommendations.

This makes CBL useful in dynamic environments. However, it faces a cold-start problem when there is no previous information about a new user or movie. The system may need demographic information, user-selected preferences, or popularity-based recommendations.

Locally Weighted Regression

LWR can also incorporate new data without retraining a global model. New observations are added to the stored dataset and can influence future local models.

For example, if a user's preferences change from action movies to documentaries, recent and similar observations can receive larger weights. However, LWR requires enough nearby labeled observations to make reliable predictions.

Adaptability

CBL is highly adaptable because it can preserve exceptional cases and use domain-specific adaptation rules. It can also explain a recommendation by showing which previous cases influenced the decision.

For example, an online shopping system might explain that a product was recommended because customers with similar purchase histories bought it successfully.

LWR adapts through locality. Nearby observations receive greater influence, while distant observations have little effect. This allows the model to respond to local trends and changing user preferences.

The bandwidth parameter controls the degree of adaptability:

- Small bandwidth: highly flexible but possibly noisy predictions.
- Large bandwidth: smoother predictions but less personalization.

Computational Efficiency

Both methods have low training costs because most computation occurs during prediction. A simple implementation may compare a query with all N training examples, requiring approximately $O(Nd)$ distance calculations, where d is the number of features.

CBL may be more efficient when it retrieves only a small number of similar cases and directly reuses their solutions. LWR generally requires additional computation because it calculates weights and solves a local regression problem for every query.

Search indexes and approximate nearest-neighbor algorithms can improve the efficiency of both approaches.

Conclusion for Question 1

Case-Based Learning is preferable when the recommendation system needs:

- Explainable recommendations.
- Rich contextual information.
- Reuse of previous solutions.
- Handling of unusual or exceptional cases.

Locally Weighted Regression is preferable when the system needs:

- A continuous prediction such as a rating.
- Local personalization.
- Adaptation to changing preferences.
- A smooth numerical estimate.

CBL focuses on retrieving and adapting previous experiences, whereas LWR focuses on building a local mathematical model from nearby examples.

2. Locally Weighted Regression vs. Radial Basis Function Networks

Locally Weighted Regression and Radial Basis Function networks both use local information. However, LWR creates a new local model for every query, while an RBF network learns a set of localized hidden units before making predictions.scribd+1

Example Dataset

Assume the following dataset represents samples from a noisy sine function:

$$y = \sin(x) + \text{noise}$$

x	y **approximately**
0	0.00
1	0.84
2	0.91
3	0.14
4	-0.76
5	-0.96
6	-0.28
7	0.66

x	y **approximately**
8	0.99
9	0.41
10	-0.54

Suppose the model must predict y when $x = 4.2$.

Locally Weighted Regression on the Dataset

LWR identifies points near $x = 4.2$, such as:

- $x = 3$
- $x = 4$
- $x = 5$
- $x = 6$

Closer points receive higher weights. The model then fits a local line or polynomial to these observations.

A common weighted error function is:

$$E(\hat{f}) = \sum_{i=1}^{N} w_i\left(x_q\right)\left(y_i - \hat{f}(x_i)\right)^2$$

where:

- x_q is the query point.
- $w_i(x_q)$ is the weight of training point i.
- y_i is the observed output.
- $\hat{f}(x_i)$ is the local model prediction.

A Gaussian weighting function may be written as:

$$w_i(x_q) = \exp\left(-\frac{\left(x_i - x_q\right)^2}{2\tau^2}\right)$$

Here, τ is the bandwidth.

Radial Basis Function Network

An RBF network uses hidden units based on radial functions, usually Gaussian functions. Each unit has a center c_i and width σ_i.

The output can be expressed as:

$$\hat{y}(x) = w_0 + \sum_{i=1}^{R} w_i \exp\left(-\frac{\| x - c_i \|^2}{2\sigma_i^2}\right)$$

For the example dataset, the network might use centers at:

$$c = [1, 3, 5, 7, 9]$$

When $x = 4.2$, the RBF units centered at 3 and 5 become strongly activated. Units centered at 1, 7, and 9 have smaller activations. The final prediction is a weighted combination of all activated basis functions.

RBF hidden units produce strong outputs near their centers and outputs close to zero when the input is far away.web.cs.hacettepe.edu

Comparison of Locality

Locally Weighted Regression

LWR determines locality separately for every query. The neighboring observations used for $x = 4.2$ may differ from those used for $x = 8.5$.

This makes LWR strongly data-driven. It can respond to changes in data density and local behavior.

RBF Networks

An RBF network learns its locality during training. The centers and widths determine which input regions each hidden unit represents.

After training, the same centers and widths are used for every query. Therefore, the model is less directly dependent on the full training dataset during prediction.

Computational Cost

Let:

- N = number of training examples.
- d = number of input features.
- k = number of local neighbors.
- R = number of RBF hidden units.

LWR Cost

A basic LWR implementation may require:

1. Searching for nearby training examples.
2. Computing distances and weights.
3. Solving a weighted regression problem.

A full search requires approximately:

$$O(Nd)$$

for distance calculations. The local regression itself may require approximately:

$$O(kd^2 + d^3)$$

depending on the implementation.

LWR therefore has low training cost but potentially high prediction cost.

RBF Network Cost

RBF networks perform more work during training. They must determine:

- Hidden-unit centers.
- Widths.
- Output weights.

After training, prediction requires calculating the activation of R basis functions. Its approximate prediction cost is:

$$O(Rd)$$

This is often faster than LWR when many predictions must be made.

Smoothness of Predictions

Criterion	**Locally Weighted Regression**	**RBF Network**
Learning style	Lazy learning.	Eager learning.
Locality	Selected separately for each query.	Fixed through learned centers and widths.
Training cost	Usually low.	Usually higher.
Prediction cost	Can be high because local regression is recomputed.	Usually low after training.

Criterion	Locally Weighted Regression	RBF Network
New data	Easily added to the stored dataset.	Usually requires retraining or updating parameters.
Memory requirement	May store most or all training examples.	Stores learned centers, widths, and weights.
Smoothness	Controlled by bandwidth and local model.	Naturally smooth with Gaussian basis functions.
Explainability	Easy to explain using nearby examples.	Less directly explainable because several hidden units contribute.
Large-scale deployment	More difficult if the dataset is very large.	More suitable after compression into a compact network.
Sparse or irregular data	Can adapt to local density.	Depends heavily on center placement.

LWR smoothness depends mainly on:

- Bandwidth.
- Number of neighbors.
- Type of local regression.
- Density of training examples.

A very small bandwidth causes the prediction to follow individual data points closely. This can produce jagged or noisy predictions. A large bandwidth creates smoother predictions but may miss local peaks and valleys.

RBF networks generally produce smooth predictions because Gaussian basis functions change continuously with the input. However, smoothness depends on the selection of centers and widths.

- Narrow Gaussian functions may create sharp local variations.
- Broad Gaussian functions create smoother but less detailed approximations.
- Poorly distributed centers may produce gaps or inaccurate regions.

Detailed Comparison

Suitable Applications

Choose LWR When:

- The data changes frequently.
- New observations must be used immediately.

- Local explainability is important.
- The dataset is small or moderate in size.
- The target function has different behavior in different regions.

Choose an RBF Network When:

- Fast prediction is required.
- Many queries must be answered.
- A compact model is preferred.
- Smooth function approximation is important.
- The data distribution is relatively stable.

Conclusion for Question 2

LWR is more flexible and directly dependent on nearby training examples. It adapts easily when new data arrive, but it may be computationally expensive during prediction.

RBF networks learn a fixed set of local Gaussian functions. They generally require more training effort but offer fast inference and smooth predictions. For a small changing dataset, LWR may be more practical. For a stable dataset requiring repeated predictions, an RBF network is often more computationally efficient.

References

1. University of Windsor, *Instance-Based Learning*, discussion of Case-Based Learning and Locally Weighted Regression.angom.myweb.uwindsor
2. Hacettepe University, *Instance-Based Learning*, discussion of local approximation and radial basis functions.web.cs.hacettepe.edu
3. Thomas, P., *Using Radial Basis Functions for Function Approximation*, discussion of the relationship between RBF networks and memory-based LWR.people.cs.umass